

Formalization, identification and decomposition of pluripotents Boolean networks

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This internship targets students with a good mathematical background and strong skills in theoretical computer science.

The proposed work builds upon previous studies conducted at the TIMC laboratory concerning Boolean networks [1-15]. While our approach primarily leans toward theoretical methodologies using non-monotonic Boolean logic and constraint-based programming, it explores how biological behaviors in individual systems can combine during evolution within larger pluripotent systems. These systems have distinct behaviors achievable through specific control sequences. Alongside the gradual embedding of biological systems during evolution, this modeling reflects how well-chosen and resilient control sequences can maintain competitive coherence within cells [16-17]. This involves cells continuing to function and survive while substantially altering their internal configurations to reach another biological state. To achieve this, we utilize discrete formalisms describing biological systems and their behaviors. Biological interaction networks and their dynamic behaviors (transition graphs) can be effectively modeled using a Boolean network formalism (interaction graphs).

Within this framework, we define a pluripotent network of dimension d as a system with an embedded transition graph containing disjoint sub-graphs, which may, in turn, contain further disjoint sub-graphs, and so on. Each sub-graph of lower dimensionality ($d-k$) can represent a transition graph of one or several Boolean networks of dimension ($d-k$). Consequently, the pluripotent network of dimension d can simulate behaviors of various smaller networks. Specific nodes in the pluripotent network serve as control nodes to select among embedded behaviors. By employing particular control sequences over multiple control nodes, a trajectory from one embedded behavior to another, through a collection of others, can be achieved. These changes must be gradual in biological systems like cells (in the context of cell functioning) or species (in the context of biological evolution) to avoid functional disruptions while transitioning to new configurations. Hence, biological systems must resolve the paradox of being highly evolvable yet robust to functional changes, while remaining in viable configurations.

Boolean networks, which well describe structures associated with biological systems and their dynamics, are conducive to our research due to their simplicity in large-scale computation. In prior studies, we demonstrated that their structure, properties, and dynamics can be precisely inferred using logic constraint-based programming languages (e.g., ASP, Answer Set Programming). This methodology encodes available knowledge, such as node interactions and their activity, as combinations of constraints translated into a logical form sent to a satisfiability solver (SAT-like). This solver produces a complete set of valid models (satisfiable sets) satisfying the logical constraints, offering a diverse collection of different yet satisfactory models. This method is not limited to the validation of individual models but furnishes a collection of models, all different but all satisfiable.

Using these formalisms and techniques our objective is to formalize the recursive decomposition of interaction graphs into disjoint sub-graphs, starting from the initial dimension to the lowest one. We aim to implement both this decomposition and the search for satisfiable sets of Boolean networks that correspond to the disjoint sub-graphs to be found. We are currently working on the formalization of the decomposition on a simple case, *i.e.* the recursive decomposition of a d -dimensional graph (associated to a d -node network, d currently limited to dimension 4). In this simple case, the student will have to perform an efficient implementation of such logical relations in a logical constraint programming language, *Answer Set Programming* (Postdam University). This will be used to solve some problems like i) finding coherent triplets of networks (N, N_1, N_2) where N is the 3D embedding network and N_1 and N_2 the embedded ones, ii) given N , finding (N_1 and N_2) coherent sets, iii) given N_1 and N_2 , finding a coherent embedded network N , *etc.* This also has to be generalized to a more general case for embedding networks of any dimension that can simulate the behaviors of numerous embedded networks.

This work includes the analysis of the populations of triplet networks that satisfy the constraints specified above, such as : (i) find a way to visualize efficiently the triplets, (ii) analyze the structural relationships

between embedded and embedding networks, (iii) study the impact of different Boolean formalisms (e.g., propositional or sign-based Boolean networks), (iv) find effective control sequences to navigate from one behavior to another, (v) identify progressive and minimal-change trajectories, (vi) design robust control sequences that tolerate errors or mutations.

Note: This work aims for a two-month internship duration (M1 level), extendable to three months. As it is a quite hard and extensible subject, it is suitable for M2 levels (6 months). This subject holds potential for a Ph.D. thesis following a M.Sc. 2.

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